

Hurford’s Constraint and Disjunctions over Speech Acts

Manfred Krifka¹

Leibniz-Zentrum Allgemeine Sprachwissenschaft (ZAS) and Humboldt-Universität zu Berlin
Berlin, Germany
krifka@leibniz-zas.de

Abstract

The paper focuses on apparent exceptions to Hurford’s generalization on disjunctions of the type *John was born in Paris or at least in France* and *John was born in France or even in Paris*. It argues that such cases are disjunctions of speech acts of different strength. For this reason, the assertion with the stronger proposition does not license the other one, nor vice versa, hence Hurford’s generalization is not violated. The paper raises the issues of disjunctions of speech acts and of assertive strength, and also reconsiders apparent exceptions of the type *Some or all of the children are sick*. Furthermore, it casts doubt on the existence of embedded scalar implicatures, and argues for a representation of speech acts in syntax and semantics.

1 Hurford’s Generalization and its Exceptions

Fifty years ago, Hurford (1974) observed that disjunctions as in (1) are infelicitous, and proposed that “the joining of two sentences by *or* is unacceptable if one sentence entails the other” (p. 410).

- (1) #*John is an American or a Californian*.

This generalization can be derived from more general pragmatic principles, like the Maxim of Manner (Grice 1975), cf. Meyer 2014 for “Efficiency”: The meaning of (1) is already expressed by the shorter *John is an American*, as it is known that everyone who is Californian also is American.

Gazdar (1979: 81), pointing out exceptions like (2), proposed that Hurford’s generalization does not apply when the entailed sentence would potentially implicate the negation of the entailing sentence (as *some were there* would implicate ‘not all were there’).

- (2) *Some or all of them were there*.

Singh (2008) and Chierchia et al. (2009, 2012) argued that cases like (2) do not constitute exceptions to Hurford’s generalization because the first disjunct is strengthened by a local scalar implicature that generates an exhaustive meaning, ‘only some were there’; as a consequence, it is not entailed by the second conjunct, *all were there*. Cases like (2) form the core argument for such embedded implicatures, often associated with a silent operator EXH (cf. Katzir & Singh 2014, Fox & Spector 2018, Repp & Spalek 2021, Hénot-Mortier 2023).

Below I will offer a different proposal. It starts with an observation by Simons (2001) and by Singh (2008), which has attracted less attention as a potential exception to Hurford’s generalization: While (3)(a) is unacceptable, as predicted by Hurford (1974), (b) is fine.

- (3) a. #*John was born in Paris or in France*.
b. *John was born in Paris or at least in France*.

I will proceed as follows. Section 2 discusses existing accounts of the contrast illustrated in (3). Section 3 proposes a new account, according to which the assertion *John was born in Paris* does

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not conversationally entail *John was born at least in France*, because while the latter is a weaker proposition, it is a stronger assertion. Section 4 widens the view to cases beyond assertions. Section 5 returns to cases like (2) and proposes an analysis in terms of weak and strong assertions as well. Section 6 considers ways to implement disjunctions of speech acts and assertions of different strength, and Section 7 concludes.

2 “At least” Exceptions: Previous Accounts

There are a few existing attempts to explain apparent exceptions to Hurford's generalization of the *at least* type exemplified by (3)(b). We will discuss them in this section.

Singh (2008) suggested, in a footnote, that the second disjunct “retracts” the first one. He states that the sentence comes with a “comma” intonation, i.e. is realized by two prosodic phrases, (*John was born in Paris*) (*or at least in France*). There are problems with this suggestion. First, it is unclear why this intonation is not able to salvage example (3)(a). Second, the comma intonation may help but does not seem to be required for (3)(b) to be acceptable. Third, true retractions as illustrated in (4)(a) cannot be applied to the case at hand, witness the oddity of (4)(b).

- (4) a. *John was born in Paris, or rather in Lyon.*
 b. *#John was born in Paris, or rather in France.*

Westera (2019, 2021) argues that *at least* in (3)(b) indicates that the two disjuncts have different value on “some scale” and relates the difference to the Question under Discussion (QuD). Building on this work, Zhang (2022) suggests that the QuD in (3)(b) is changed from ‘In which city was John born?’ to ‘In which state was John born?’. As the disjuncts answer different QuDs, they do not stand in an entailment relation to each other. I think that an explanation along this line is problematic, as we can fix the QuD explicitly, and still can recreate the contrast in (3):

- (5) A: *In which city was John born?*
 B: *He was born in Paris or #in a city in France / at least in a city in France.*

According to Simons (2001), the individual clauses in a regular disjunction should give distinct, non-overlapping answers to the same QuD. For disjunctions with the second disjunct marked by *rather* or *at least*, Simons argues that they have a “metalinguistic” nature and “indicate a particular relation between speech acts” in which “the second ‘disjunct’ is to fill the same ‘discourse slot’ as the first, and thus in some sense replaces that act” (p. 613). The second disjunct requires some “qualifying expression” such as “*at least, rather, even, I should say and should I say?*” Simons observes that such disjunctions do not allow for *either—or* constructions (6)(a), that they are not symmetric (b), and that *or* can be eliminated (c):

- (6) a. *#Either George lives in London, or at least he lives somewhere in England.*
 b. *#George lives (at least) somewhere in England, or in London.*
 c. *Georges lives in London. At least, he lives somewhere in England.*

The proposal that I will put forward is inspired by Simon's impression that the disjunctions combine different speech acts. I will develop the idea that (3)(b) is felicitous because the disjunction can scope over two assertions, where the first assertion does not license the second, as the second is a stronger assertion, as indicated by *at least*. Without *at least* as in (1), the assertions are of the same strength; consequently the first disjunct licenses the second one, and the configuration is ruled out by the pragmatic factors that are responsible for Hurford's generalization.

3 Disjunction of Assertions and Assertive Strength

I propose that disjunctions that appears within assertions can be read as disjunctions of assertions. Assuming the commitment account of assertions (cf. Brandom 1994, Krifka 2015, Geurts 2019, Shapiro 2020), I represent the assertion by speaker S of proposition Φ as “ $S \vdash \Phi$ ”, that S vouches for the truth of Φ . Hence, I propose that the assertion by S in (7)(a) has, in addition to the propositional assertion (7)(b), a reading with a disjunction on the speech act level, (7)(c).

- (7) a. S: *John was born in Paris or in Lyon.*
 b. $S \vdash [\text{‘John was born in Paris’} \vee \text{‘John was born in Lyon’}]$
 c. $[S \vdash \text{‘John was born in Paris’}] \vee [S \vdash \text{‘John was born in Lyon’}]$

Readings involving disjunctions over speech acts are often disfavored because they lead to ambiguous discourse states. After (7)(c), speaker S is either committed to ‘John was born in Paris’ or to ‘John was born in Lyon’, a commitment that is difficult to keep track of in discourse. Therefore, disjunctions of speech acts were supposed to be ruled out. For example, Krifka (2001) derived the fact that quantification into constituent questions, as in *Which dish did every guest bring to the party?* is possible only with universal quantifiers like *every guest*, i.e. with generalized conjunctions, but not with other quantifiers like *most guests* that are also based on disjunctions. For (7)(a), this means that the reading with a proposition-scope disjunction (7)(b) is strongly favored, as this results in a non-ambiguous commitment of S to a (disjunctive) proposition.

Hurford's generalization, originally based on an entailment between propositions, can be recast for disjunctions of assertions. A speaker that commits to a proposition Φ is also obliged to commit to a proposition Ψ , if Φ entails Ψ in a transparent manner (cf. commitment inheritance or entitlement, Brandom 1994). Hence the commitment to Ψ is redundant, resulting in the oddness of (1) and (3)(a). This observation may seem irrelevant if disjunctions of assertions are disfavored anyway, for the reason given above. But now we can develop an argument why (3)(b) is good.

The crucial property of (3)(b) is that the second disjunct contains the modifier *at least*. Such “superlative” modifiers have been argued to be interpreted not on the propositional level, but on the level of epistemic operators (Geurts & Nouwen 2007) or speech acts (Cohen & Krifka 2014). Assuming that *at least* is focus-sensitive and hence relates to alternatives, I propose that it can modify assertions, resulting in a commitment to the asserted proposition with a strength that is different from the one to the alternative propositions. In disjoint assertions of the type (3)(b), the proposition of the first disjunct is an explicitly mentioned alternative to the proposition of the second disjunct. The proposition ‘John was born in France’ is logically weaker than ‘John was born in Paris’, and for this reason can be committed to and hence asserted more strongly. (Note that there are cases that are not based on logical entailment but on argumentative weight, as in *Mary is a full professor or at least an associate professor*, cf. Ducrot 1973). Thus, the reading of (3)(b), with disjunction scoping over the commitment, has the following interpretation:

- (8) S: $[S \vdash \text{‘John was born in Paris’}] \vee [S \vdash_{\text{strong}} \text{‘John was born in France’}]$

The subscript indicates that the second commitment is stronger than the first. That speech acts may come with various strengths has been proposed before, e.g. Searle & Vanderveken (1986), who include a 5-level strength parameter, and Greenberg & Wolf (2018), who state that assertions come with different degrees. Different strengths of assertions have typically been discussed in terms of mitigation of assertions, cf. McDowell (1987), Sbisà (2001), and Incurvati & Schlöder (2019). We make the plausible assumption that $S \vdash \Phi$ does not oblige S to $S \vdash_{\text{strong}} \Phi$; also, $S \vdash \Phi$ does not oblige S to $S \vdash_{\text{strong}} \Psi$ even if Φ transparently entails Ψ . Note that under this analysis, cases like (3)(b) do not violate Hurford's generalization and are predicted to be felicitous.

We predict that other modifiers of assertive commitment strength have similar effects. *Definitely* and *in any case* occur with the less specific proposition and should be classified as strengtheners, like *at least*, cf. (9)(a). This also holds for *certainly*, which makes the assertion of a proposition easier to defend, hence assertable under more circumstances (cf. Lassiter 2016, Yatsushiro et al. 2021). Assertive commitments with different strength do not require a weakening of the proposition but can also involve hedging by weakening the evidential backup, cf. (9)(b).

- (9) a. *John was born in Paris or definitely / in any case / certainly in France.*
 b. *John was born in Paris, or at least he told me so.*

Simons (2011), Westera (2019) and Zhang (2022) pointed out that the particle *even* can also lead to apparent exemptions from Hurford's generalization, cf. (10)(a). In contrast to *at least*, *even* occurs with the more specific proposition, which I take to flag a weaker commitment in the context of a disjunction with an alternative proposition, leading to the analysis (10)(b). Classifying the second assertion as weaker may seem counterintuitive, as *even* seems to mark extreme alternatives. However, this applies to the proposition in comparison with its alternatives, and not to the assertive strength itself (cf. the fact that *perhaps* and *even* happily cooccur).

- (10) a. *John was born in France or even in Paris.*
 b. $S \vdash \text{'John was born in France'} \vee S \vdash_{\text{weak}} \text{'John was born in Paris'}$

The order of disjuncts is relevant; assertions like $\# \text{'John was born at least in France or in Paris'}$ and $\# \text{'John was born even in Paris or in France'}$ are infelicitous. This order-sensitivity (cf. also Tomioka 2021) holds for focus-sensitive operators in general, as explicitly mentioned alternatives have to be mentioned before the operator in order to be understood as a relevant alternative (e.g. *I met John and even Bill* vs. $\# \text{'I met even Bill and John'}$).

4 Beyond simple assertions

Operators like *at least* and *even* also occur in disjunctions of other speech acts than assertions (cf. Ciardelli & Roelofsen 2017 for questions), where they can abide by Hurford's generalization due to similar asymmetries as with assertions. With questions like *Was John born in Paris or at least in France?* the second disjunct is a more likely answer. With imperatives like *Go to Paris or at least to France!* the second disjunct is a stronger command.

Zhang (2022) has observed disjunctions in conditionals as in *Mary will date John if he was born in Paris or at least in France*. Such cases can be explained by assuming conditional assertions (Krifka 2019). The example can be analyzed as $[\text{'J was born in Paris'} \rightarrow S \vdash \text{'Mary will date John'}] \vee [\text{'J was born in France'} \rightarrow S \vdash_{\text{weak}} \text{'Mary will date John'}]$; notice that the strength of the second commitment is now weaker, as *at least* occurs in a downward-licensing position. In contrast to *definitely*, which is not focus-sensitive, *at least* has a strength parameter that can be set freely to a lower or higher degree in order to satisfy Hurford's generalization.

The strength parameter of *at least* is not even restricted to speech acts but can relate to other strength-sensitive operators, like propositional attitudes. Consider the sentence *Mary believes that John was born in Paris or at least in France*; in addition to a speech act-related interpretation, $[S \vdash \text{'M BEL J was born in Paris'}] \vee [S \vdash_{\text{strong}} \text{'M BEL J was born in France'}]$, there is an interpretation in which *at least* expresses a stronger belief, $S \vdash [\text{'M BEL J was born in Paris'} \vee \text{'M BEL}_{\text{strong}} \text{J was born in France'}]$. In this interpretation, S vouches for the proposition that Mary has a belief of regular strength that John was born in Paris, and a stronger belief that John was born in France. This description of Mary's belief is not redundant, as we cannot infer from Mary's regular belief in the first proposition to Mary's strong belief in the second.

In concluding this section, I want to point out that in all these cases of apparent violations to Hurford's generalization, there is an additional pragmatic reason for the expression of two disjuncts. With presenting two disjuncts, sentences like (3)(a) and (9)(a) introduce two concepts (Paris and France) and draw attention to two propositions (that John was born in Paris, that John was born in France). This non-truthconditional aspect in Hurford disjunctions was pointed out by Westera (2019). However, notice that this is not sufficient to explain why apparently redundant disjunctions are pragmatically acceptable, as we would then predict that cases like (3)(a) should be acceptable as well, as they too draw attention to different propositions.

5 “Some/All” Exceptions: An Alternative Account

For the apparent exceptions to Hurford's generalization considered so far, it was essential that one disjunct (the second) had an explicit marker, like *at least* or *even*, that arguably indicated a different strength from the other (the first). Disjunctions without this marker, as in (3)(a), cannot not simply coerce the second disjunct into a stronger or weaker reading.

Turning to apparent exceptions of the *some or all* type like (2) and (11)(a), we see that such explicit markers are not necessary; we do not have to insert *only* before *all* in (11). In fact, there is a slight degradation with overt *only*; on Google Ngrams the string *only some or all* occurs more than 1000 times less frequently than *some or all*, and seems largely confined to texts in logic, computer science, and linguistics.

- (11) a. *John read some (most, several) or all novels by Dickens.*
 b. *John read three or seven novels by Dickens.*
 c. *John read ‘Oliver Twist’ or ‘Bleak House’, or both of these novels.*

We observe that exceptions of the *some or all* type can be reproduced with other items, in particular number words as in (11)(b); note that the proposition ‘John read 7 novels by Dickens’ entails the proposition ‘John read 3 novels by Dickens’. (Cases like *John read three or four novels by Dickens*, spoken with one prosodic phrase over the disjunction, have to be treated separately, as approximate number words on a scale where they do not compete with *three* or *four*).

There are several proposals to explain apparent exceptions to Hurford's generalization of the *some or all* type (cf. Marty & Romoli 2022 for a critical overview). I would like to propose that these cases can be subsumed under the exceptions involving different assertive strengths if we assume that the second disjunct makes a weaker assertion, as in (12). This is similar to our analysis of cases with *even* as in (10). Note that we could insert *even* before *some* and *seven* in (11)(a,b); in fact, on Google Ngrams *some or even all* only occurs only 20 times less frequently than *some or all*.

- (12) a. $S \vdash \text{‘John read some of the novels by D’} \vee S \vdash_{\text{weak}} \text{‘John read all of the novels by D’}$
 b. $S \vdash \text{‘John read 3 novels by D’} \vee S \vdash_{\text{weak}} \text{‘John read 7 novels by D’}$
 c. $S \vdash \text{‘John read OT or BH’} \vee S \vdash_{\text{weak}} \text{‘John read OT and BT’}$

The question arises why cases like (11) do not require an explicit marker of assertive strength of the second disjunct, in contrast to (3), (9) and (10). A plausible reason is that these cases involve lexical alternatives, or Horn scales; hence purely linguistic competence will result in the recognition of differences in assertive strength; in contrast, alternatives like *Paris / France* require access of encyclopaedic knowledge. An additional difference is that quantifiers like *some*, *all*, numerals like *three*, *seven* and Boolean connectives like *or*, *and* (or *both*) form total (linear) relations, in contrast to part relations or hyponymy relations like *Paris*, *France* or *beetle*, *insect*. The lexical nature of the alternatives, perhaps in addition to their linear semantic order, reduces the need to mark the non-default strength of the assertion explicitly. Merin (2003), without referring to Hurford's

generalization, discusses the use of *if not* in such cases, as in *some if not all* and *three if not four*, and develops an analysis in terms of act-based relevance similar to the one proposed here.

The two competing analyses both make use of the notion of alternatives; one understands *some or all* as ‘only some or all’, the other as ‘some or even all’. There is a structural difference: In the first paraphrase, the alternative-sensitive operator, *only* or EXH, occurs before the disjunct that mentions the alternative; in the second, the operator *even* occurs after the disjunct that mentions the alternative. We have seen that explicitly mentioned alternatives should occur before the alternative-sensitive operator; this is a further argument for the analysis developed here.

6 Implementing Speech Act Strength and Disjunction

In sections 3 and 5 I sketched a theoretical account from which apparent exceptions to Hurford's generalization lose their exceptional status. It makes two essential assumptions: We can disjoin speech acts, and we have assertions of different strength. In the remainder of this short paper, I will point to research that show how these notions can be formally implemented, and tested.

Disjunctions of speech acts can be handled in the framework of Commitment Spaces (CS) of Krifka (2015), where this operation is central for alternative questions and constituent questions. The CS framework comprises the common ground at a particular state of conversation, the root, together with its possible continuations. The root is ideally a single commitment state, but it can also be ambiguous, containing more than one commitment state (cf. Kamali & Krifka 2020). Update of the root $\{c\}$ with a commitment $S \vdash \Phi$ to $\{c + S \vdash \Phi\}$ triggers further update with the asserted proposition Φ to $\{c + S \vdash \Phi + \Phi\}$ in case the addressee does not object (cf. Krifka 2022). Update of a CS with root $\{c\}$ with a disjunction of assertions, like $S \vdash \Phi \vee S \vdash \Psi$, results in an ambiguous root $\{c + S \vdash \Phi, c + S \vdash \Psi\}$, which should be avoided, as it creates a discourse state at which it is unclear whether Φ or Ψ can be assumed. However, in the case of disjunctions like (8), of the form $S \vdash \Phi \vee S \vdash_{\text{strong}} \Psi$, where Φ transparently entails Ψ , the resulting discourse state allows for the introduction of the more general proposition Ψ , as it is supported by both $S \vdash_{\text{strong}} \Psi$ and, indirectly, by $S \vdash \Phi$. Hence the resulting discourse state is considerably less ambiguous, as it allows for assuming Ψ in any case. In addition, this discourse state gives partial and weaker support for Φ , similar to ‘ Ψ and perhaps Φ ’, and makes the proposition Φ salient so that it can be taken up later (cf. Westera 2019). A similar argument can be developed for disjunctions like (10) and (12).

The cases we have considered all involved disjunctions of sub-clausal constituents, such as *in Paris or at least in France* for (3)(b), which appears at odds with the idea that they indicate disjunctions of assertions. However, we can assume that redundant material is not phonologically realized, hence (3)(b) is derived by deletion, *John was born in Paris or ~~John was born~~ at least in France*. Alternatively, we can explore the idea that there is an interpretation to the disjunctive phrase *in Paris or at least in France* that takes a speech act with a trace as an argument, like $\lambda x \lambda c [c + S \vdash \text{‘John was born in } x\text{’}]$. Such quantifications into speech acts have been proposed for alternative questions and constituent questions in Krifka (2015).

As a final point I want to raise the notion of strength. In the commitment account of assertions, assertive strength can be operationalized by the level of warrant with which the speaker backs up a commitment to the asserted proposition. This idea was introduced by Ch. S. Peirce in terms of the “esteem”, or reputation, of the speaker that may be affected if the asserted proposition turns out to be false and the speaker does not have a sufficient excuse (cf. Tuzet 2006). In general, stronger assertions result in an increased threat to one's reputation than weaker ones, where strength can be mitigated or increased by epistemic and evidential operators and by direct modifications of commitments (cf. Krifka & Modarresi 2023). In accounts of assertions that work with the notion of expressed beliefs, one would have to assume different belief strengths.

7 Conclusion

I have proposed a novel explanation of exceptions to Hurford's generalization like (3)(b) that have not been given a satisfying explanation so far, and tentatively extended this to better-known exceptions (2) that have been explained differently, by embedded exhaustification. Our approach also throws light on the possible scope of disjunction over speech acts and about the representation of speech acts in grammar. It also provides new evidence for the graded nature of assertive commitments and the mitigation and strengthening of speech acts.

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**Leibniz-Zentrum
Allgemeine Sprachwissenschaft**



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Manfred Krifka

Leibniz-Zentrum Allgemeine Sprachwissenschaft (ZAS) Berlin

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Hurford's Constraint and its exceptions

■ Hurford's Constraint

— Hurford 1974:

- *#John is an American or a Californian.*

— Generalization:

“the joining of two sentences by or is unacceptable if one sentence entails the other”

- *John is a Californian* (second disjunct)
Every Californian is an American. (presupposed encyclopedic knowledge)

John is an American. (first disjunct)

— Explanations (Grice's Maxim of Manner, e.g. Meyer 2014, “Efficiency”):

The entailed sentence is redundant,

- Here: suggests that Californians are not Americans, contrary to encyclopedic knowledge

Exceptions to Hurford's Generalization

■ Some/all exceptions:

- Gazdar 1979: Hurford's Generalization does not apply when the entailed sentence would potentially implicate the negation of the entailed sentence.

- *Some or all of them were there.*

Note: *Some of them were there* would implicate 'Not all of them were there'

- Making Hurford's Generalization exceptionless:

- The conjunct *some of them were there* actually "means": 'Some but not all of them were there', no violation of Hurford's generalization
- by a scalar implicature triggered within the disjunction
- or by a silent EXH operator:
Singh 2008, Chierchia et al. 2009, 2012, Katzir & Singh 2014, Fox & Spector 2018, Hénnot-Mortier 2023, ...

■ At-least exceptions:

- Simons (2001),
apparent exception to the "distinctness condition" / "non-vacuity principle"
- *George lives in London, or at least he lives somewhere in England.*

Previous explanations of At-least exceptions

■ Simons 2001:

- Disjunctions should give distinct, non-overlapping answer to the same QuD
- Second disjunct with *at least* or *rather* : metalinguistic, replacement of first speech act.
 - *George lives in London, or at least he lives somewhere in England.*
 - *George has bought a car, or rather he has bought a brand new Ferrari.*

■ Singh 2008, footnote:

- *John was born in Paris, or at least in France*
- No actual disjunctions, but “retractions” whereby speaker “weakens initial assertion”
- Comma intonation (i.e. phonological phrase boundary)
 - *(John was born in Paris)(or at least in France)*

■ Problems:

- “Comma intonation” does not seem to be required
- Regular retractions do not need “at least” in second disjunct
 - *John was born in Paris, or rather (# at least) in Lyon.*
- True retractions cannot be applied to the core case:
 - *John was born in Paris # or rather in France.*

Previous explanations of At-least exceptions

■ Westera 2019, 2022

- *John was born in Paris or at least in France.*
- The two disjuncts have a “different value” on “some scale” relating to the QuD
- They differ non-truthconditionally in making different propositions salient

■ Zhang 2022

- The two disjuncts answer different QuDs,
‘In which city was John born?’ vs. ‘In which state was John born?’
- Hurford’s generalization applies only when disjuncts answer the same QuD
- Problem: Violation / Non-violation appears also when QuD is fixed:
 - A: *In which city was John born?*
 - B: #*John was born in Paris or in a city in France.*
 - B: *John was born in Paris or at least in a city in France.*

A new account: Commitment strength



■ Commitment accounts of assertion

- In an assertion, a speaker vouches for the truth of the asserted proposition: (Peirce ca. 1905, Brandom 1994, Geurts 2019; Shapiro 2020, ...)
- Krifka 2015: $S \vdash \Phi$, “speaker S is publicly committed to proposition Φ ”

■ Commitment licensing

- Committing to certain propositions may commit a speaker to other propositions
- A commitment $S \vdash \Phi$, when $\Phi \Rightarrow_{\text{transparent}} \Psi$, licenses a commitment $S \vdash \Psi$, i.e. a speaker that committed to Φ also has to commit to, and defend, Ψ
- The commitment $S \vdash \Phi$, where $\Phi \Rightarrow_{\text{transparent}} \Psi$, makes the commitment $S \vdash \Psi$ redundant
- Explains Hurford: # $S \vdash \text{‘John was born in Paris’} \vee S \vdash \text{‘John was born in France’}$

■ Commitment strength

- Assertion strength: Searle & Vanderveken 1985, Greenberg & Wolf 2018, Incurvati & Schloeder 2019, McDowell 1987, Sbisà 2001, Krifka 2023, ...
- Related to potential reputation loss of speakers: Krifka & Modarresi 2023

■ Commitment strength and commitment licensing

- A commitment $S \vdash \Phi$ does not license a commitment $S \vdash_{\text{strong}} \Phi$ (but vice versa)
- A commitment $S \vdash \Phi$, when $\Phi \Rightarrow_{\text{transparent}} \Psi$, does not license a commitment $S \vdash_{\text{strong}} \Psi$
- Saving Hurford: $S \vdash \text{‘John was born in Paris’} \vee S \vdash_{\text{strong}} \text{‘John was born in France’}$

Marking of assertive strength: *at least*

■ Deriving strength differences for *at least*

- Not a propositional modifier, but related to epistemics: Geurts & Nouwen 2007
- Modifier on the speech-act level: Cohen & Krifka 2014
- Sensitive to alternatives of different logical strength or argumentative value (Ducrot 1973):
 - *John ate at least three apples.* ‘J ate 1 apple’, ‘J ate 2 apples’, ‘J ate 3 apples’, ‘J ate 4 apples’, ...
 - *Ann is an associate professor.* ‘Ann is an assistant professor.’, ‘...associate...’, ‘...full...’
- Comparison of logical/argumentative strength and possible assertive strength

	Content	Assertion	Possible assertive strength
• $S \vdash$ ‘John ate 3 apples’	weaker	less risky	stronger
• $S \vdash$ ‘John ate 4 apples’	stronger	more risky	weaker

— Assertion *At least* Φ

- indicates the presence of alternatives Ψ
- interpreted as $S \vdash_{St1} \Phi$ and licensing $S \vdash_{St2} \Psi$, where **St1** is **stronger** than **St2**.

■ Evading violation of Hurford’s generalization:

- Explicit mentioning of an alternative
 - S: *John was born in Paris or at least in France.*
 - $S \vdash_{St2} \text{‘J was born in Paris’} \vee S \vdash_{St1} \text{‘John was born in France’}$, with **St1** **stronger** than **St2**

Assertive strength: *definitely* and *even*

■ Other markers of strong assertions

- Apparent exceptions to Hurford's generalization
 - S: *John was born in Paris or definitely / in any case / certainly in France.*
 - $S \vdash \text{'John was born in Paris'} \vee S \vdash_{\text{strong}} \text{'John was born in France'}$

■ *Even* as marking of weak assertions

- Apparent exceptions to Hurford's generalization: Simons 2001
 - S: *John was born in France or even in Paris*
 - $S \vdash \text{'John was born in France'} \vee S \vdash_{\text{weak}} \text{'John was born in Paris'}$
 - Commitment $S \vdash_{\text{weak}} \Phi$, where $\Phi \Rightarrow_{\text{transparent}} \Psi$, does not license $S \vdash \Psi$
- Assertion *Even* Φ
 - indicates the presence of alternatives Ψ
 - is interpreted as $S \vdash_{\text{St1}} \Phi$ and implicating $S \vdash_{\text{St2}} \Psi$, where **St1** is **weaker** than **St2**.
- Evidence for compatibility with weakening:
 - *John was born in France or perhaps even in Paris.*

Some / all and Hurford's Generalization

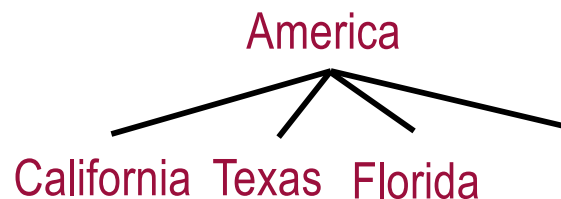


- Explanation of some/all exceptions
 - S: *Some or all of the children are sick.*
 - S: *John read three or six novels by Dickens.*
 - S: *John read Oliver Twist or Bleak House or both of these novels.*
- Explanation by “only” interpretation of first conjunct (Singh 2008, ...)
 - ‘Only some or all of the children are sick.’
 - ‘John read only three or six novels by Dickens’
 - ‘John read only Oliver Twist or Bleak House, or both of these novels’
- Alternative explanation by “even” interpretation of second conjunct
 - ‘Some or even all of the children are sick.’
 $S \vdash \text{‘Some children are sick’} \vee S \vdash_{\text{weak}} \text{‘All children are sick’}$
 - ‘John read three or even six novels by Dickens.’
 $S \vdash \text{‘John read three novels by Dickens’} \vee S \vdash_{\text{weak}} \text{‘John read six novels by Dickens’}$
 - ‘John read Oliver Twist or Bleak House or even both of these novels’
 $S \vdash \text{‘John read Oliver Twist or Bleak House’} \vee S \vdash_{\text{weak}} \text{‘John read Oliver Twist and Bleak House’}$
- An argument for the alternative interpretation
- Google ngram frequency: *some or even all* about 20 times as frequent as *only some or all*

Some / all and Hurford's generalization



- Dispensability of strength indicators
 - How to account for the following contrast?
 - *John was born in France or #in Paris / or even in Paris.*
 - *John read some or all / or even all of the novels by Dickens.*
 - *John read three or six / three or even six novels by Dickens.*
- Encyclopaedic vs. lexicalized alternatives
 - Encyclopedic alternatives (mereological, taxonomical) require explicit marking by *even*
 - *John lives in America or #(even) in California.*
 - [Tasting Bordeaux wines:] *This is a Pomerol or #(even) a Chateau Pétrus.*
 - Lexicalized alternatives (Horn scales) do not require explicit marking by *even*
〈*some, all*〉, 〈... *three, four, five, six, ...*〉, 〈*or, and/both*〉
 - Logical relations of Horn scales are part of the linguistic knowledge,
no need for explicit support by a particle that signals logical / argumentative relations
- Partially ordered vs. totally ordered alternatives



all
most
some

five
four
three

and/both
or

Some additional points



■ Order matters

— Different judgments with different sequential orders

- # *John was born at least in France or in Paris.*
- To impute a strong assertion, *at least* requires access to alternatives, hence *in Paris* must come first

■ Creation of coarse-grained scalar items

— Examples, with prosodic phrasing:

- *John has read (three or four) / (some or all) of the novels by Dickens.*
- *John has read (some or all) of the novels by Dickens.*

— Vague interval numerals:

- *three or four, ten or twelve, not: thirteen or seventeen*
- German: *Hans hat drei, vier Romane von Dickens gelesen.*

— Vague quantifier words:

- *some or all* – no implicature to ‘not all’

■ Relevance for QuD (cf. Simons 2001)

— The two disjuncts answer the same QuD – pace Zhang 2022

- [Does John know how to surf?] *John lives in America or even in California* / #*Kansas*.
- [Is John religious?] *John lives in America or even in Kansas* / #*California*.

Exploitation of strength differences

■ Strength in other speech acts

— Commands

- S to A: *Go to Paris or at least to France.*
 $\text{COMMAND}(A \text{ goes to Paris}) \vee \text{COMMAND}_{\text{strong}}('A \text{ goes to Paris})$

— Questions

- S to A: *Was John born in Paris or at least in France?>*
 S checks whether $A \vdash 'John \text{ was born in Paris}'$ or $A \vdash_{\text{strong}} 'John \text{ was born in France}'$

■ Strength in embedded clauses

— Strength distinctions indicated by *at least* can be exploited below the speech-act level

- *Mary believes that John was born in Paris or at least in France.*
- $[S \vdash 'M \text{ BEL } J \text{ was born in Paris}'] \vee [S \vdash_{\text{strong}} 'M \text{ BEL } J \text{ was born in France}']$
- $S \vdash ['M \text{ BEL } J \text{ was born in Paris}' \vee 'M \text{ BEL}_{\text{strong}} J \text{ was born in France}']$

Exploitation of strength differences

■ Strength in restrictors

- Observation by Zhang (2022)
 - *Mary will date John if he was born in Paris or at least in France.*
- Intuitively: The assertion that Mary will date John can be made more securely when John was born in Paris than when (it is only known that) he was born in France.
- Analysis as conditional assertions (Krifka 2019)
 - $['J \text{ was born in Paris}' \rightarrow S \vdash \text{'Mary date John'}] \vee ['J \text{ was born in France}' \rightarrow S \vdash_{\text{weak}} \text{'Mary date John'}]$
 - Note: *at least* signals weaker assertion as it occurs in the downward-entailing restrictor
 - Different from *definitely*, *in any case* which are fixed to indicate strength and cannot be embedded
- Strength differences in nominal restrictors, but apparently only in parenthetical constructions
 - *Every person in France, or at least in Paris, has been to Notre Dame cathedral.*
 - $S \vdash \text{'Every person in France has been ...'} \vee S \vdash_{\text{strong}} \text{'Every person in Paris has been...}'$

■ Strength in Hurford conditionals

- Observation by Mandelkern & Romoli (2018), and exception
 - # *If John is not in Paris, he is in France.*
 - *If John is not in Paris, he is at least / in any case / definitely in France.*
- Intuitively: The assertion $S \vdash \text{'John is in Paris'}$ cannot be made or is retracted, but the assertion $S \vdash_{\text{strong}} \text{'John is in France'}$ is made.
- Requires analysis of conditionals as conditional assertions, cf. Krifka (2019)

But can speech acts be disjoined?

- Argument against disjunction of speech acts: Krifka 2001
- Quantification into question acts only for universal quantifiers, as they are generalized conjunctions
 - *What did every guest bring to the party?*
'What did John bring, AND what did Mary bring, AND what did Sue bring to the party?'
 - *#What did most guests bring to the party?*
'What did John bring, and what did Mary bring, OR what did John bring, and what did Sue bring, OR what did Mary bring, and what did Sue bring?'
- Disjunctions of speech acts are avoided because they create an unclear commitment
 - *John was born in Paris or in Lyon.*
 - S: ASSERT('John was born in Paris') $\vee$ ASSERT('John was born in Lyon')
A can neither assume 'John was born in Paris', nor 'John was born in Lyon'
 - rather: ASSERT('John was born in Paris' $\vee$ 'John was born in Lyon')
A can assume 'John was born in Paris' $\vee$ 'John was born in Lyon'

Disjunctions of speech acts

- A closer look at speech act disjunction
 - Speech acts as updates of the common ground (CG)
 - At c : S to A: *'it is raining'* = $c + \text{St} \text{'it is raining'}$ = $c \cup \text{St} \text{'it is raining'}$,
 - if A does not object, further update to $c \cup \text{St} \text{'it is raining'} \cup \text{'it is raining'}$
 - Conjunction of assertions as subsequent update
 - At c : S to A: *'it is raining, and it is freezing'* = $c + \text{St} \text{'it is raining'} + \text{St} \text{'it is freezing'}$
 - Disjunction of assertions
 - At c : S to A: *'It is raining, or it is freezing'* = $\{c + \text{St} \text{'it is raining'}, c + \text{St} \text{'it is freezing'}\}$
 - ambiguous common ground, licenses neither update with *'it is raining'* nor with *'it is freezing'*
 - update with disjunctive proposition instead: $c + \text{St} [\text{'it is raining'} \vee \text{'it is freezing'}]$
- When disjunction of assertion is okay:
 - Logical relation between assertions
 - At c : S to A: *John was born in Paris or at least in France.*
 - = $\{c + \text{St} \text{'John was born in Paris'}, c + \text{St}_{\text{strong}} \text{'John was born in France'}\}$
 - Licenses further update with *'John was born in France'*, with weaker support: *'... was born in Paris'*
 - Cf. Krifka 2015, Kamali & Krifka 2020 for CGs as sets of information states

Summing up

- Hurford's generalization:
 - Disjuncts should not entail each other: #*John was born in Paris or in France*.
- Apparent exceptions to Hurford's generalization:
 - Some or all – exceptions
 - *Some or all of the children are sick.*
 - At least / even exceptions:
 - *John was born in Paris or at least in France.*
 - *John was born in France or even in Paris.*
- Proposed explanation
 - Disjunction on the speech act level
 - Different assertion strength makes disjunctions compatible with Hurford's generalization
 - $S \vdash$ 'John was born in Paris' $\vee$ $S \vdash_{\text{strong}}$ 'John was born in France'
 - $S \vdash$ 'John was born in France' $\vee$ $S \vdash_{\text{weak}}$ 'John was born in Paris'
 - Different strength does not have to be indicated explicitly with lexical scales
 - $S \vdash$ 'Some of the children are sick' $\vee$ $S \vdash_{\text{weak}}$ 'All of the children are sick'
- Relevance:
 - Speech acts in syntax and semantics; operators scope over speech acts
 - Disjunction over speech acts
 - Perhaps there is no need to assume EXH / embedded scalar implicature

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